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Movie Recommendation System Development Presentation

Group 8
Advanced Marketing Analytics

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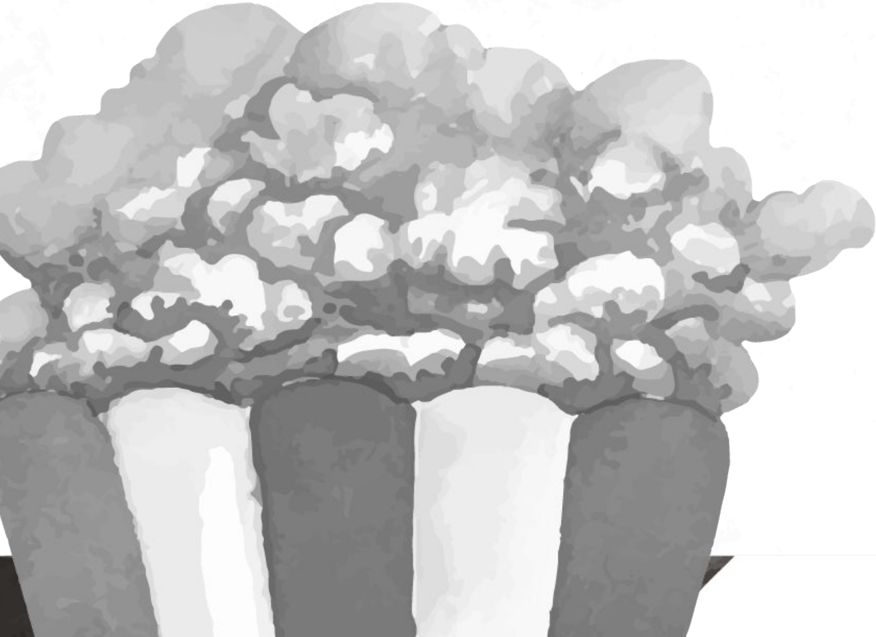
01

Introduction

Background and Data Cleaning



Background & Dataset Description

Research Question:
By leveraging the use of structured and unstructured data, how can we predict individuals' preferences and recommend movies for the individual?

Objective:
Develop a model which could find individuals' preferences based on similarities between users and between movie items.

About the datasets:

- df_ratings: contains users' historical movie ratings
- df_meta: contains metadata information about movies

	df_meta: metadata_updated.json	**df_ratings:** ratings.json
Variables	title	item_id
	directedBy	user_id
	starring	rating
	avgRating	
	imdbId	
	item_id	

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Data Cleaning & Visualization

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Step 1:
Check whether there is any null data

```
df_ratings.isnull().sum()
```

```
item_id    0
user_id    0
rating     0
dtype: int64
```

Step 2:
Aggregate to remove duplicated data

```
# Aggregate to remove duplicate ratings
df_ratings = df_ratings.groupby(['user_id', 'item_id'],
                                as_index=False).agg({'rating': 'mean'})
```

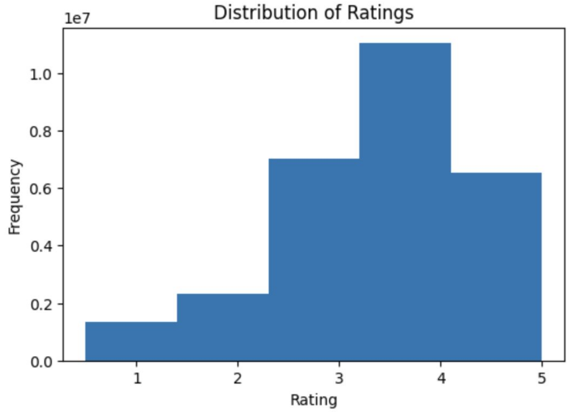
Step 3:
Data info & Describe Data

```
df_ratings.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 28249191 entries, 0 to 28249190
Data columns (total 3 columns):
 #   Column   Dtype
---  ------   -----
 0   user_id  int64
 1   item_id  int64
 2   rating   float64
dtypes: float64(1), int64(2)
memory usage: 646.6 MB
```

Step 4:
Graph the distribution of ratings

```
plt.figure(figsize=(6, 4))
plt.hist(df_ratings['rating'], bins=5)
plt.title('Distribution of Ratings')
plt.xlabel('Rating')
plt.ylabel('Frequency')
plt.show()
```



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02

Methodology

Various Modeling Approaches of
Collaborative Filtering

Explored Approaches – Collaborative Filtering

01

Initial Approach: Singular Value Decomposition (SVD)

Goal: Decompose user-item interaction matrix to predict missing values.

Challenge: Impractical for large-scale applications.

02

Alternative Approach: Alternating Least Squares (ALS)

Advantages: Highly efficient and uses implicit feedback to uncover hidden user-item patterns.

Challenge: Unable to map user indices back to original user IDS accurately.

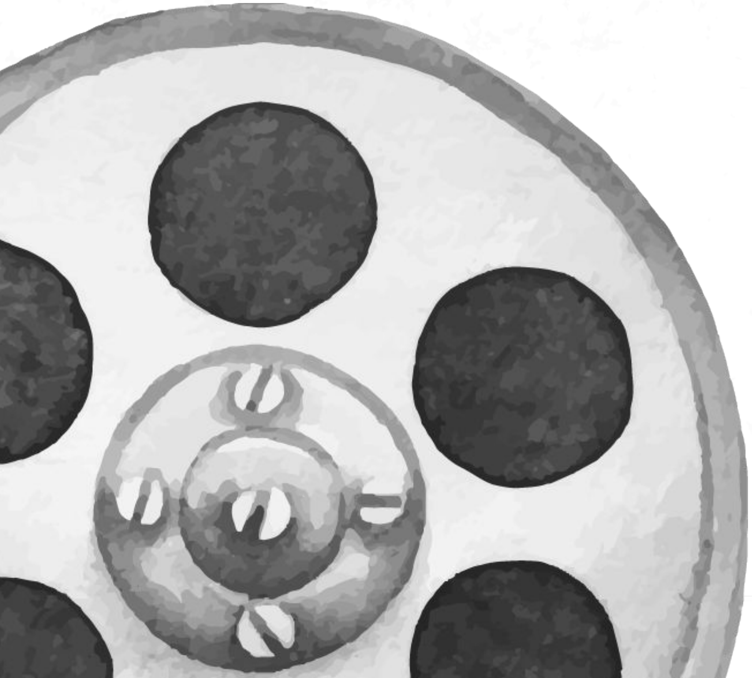
03

Neural Collaborative Filtering (NCF)

03

Findings

NCF Model with Tensorflow

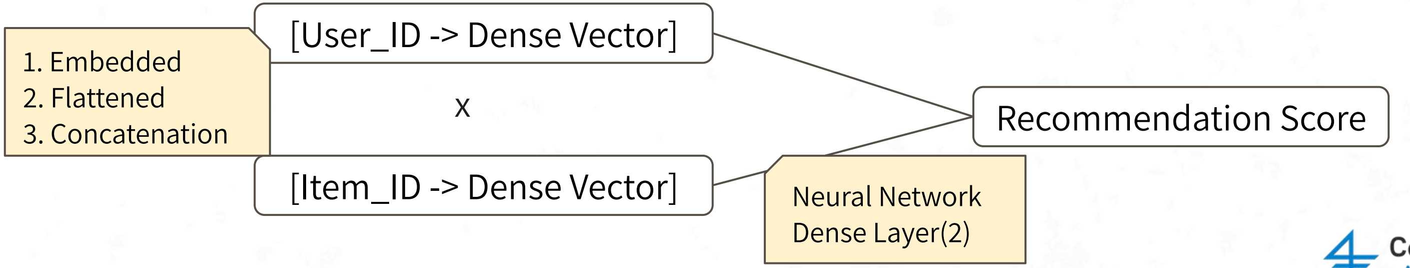


Neural Collaborative Filtering (NCF) with Tensorflow

Neural Collaborative Filtering (NCF) is an recommendation algorithm that **combines collaborative filtering with deep learning** to model user-item interactions.

TensorFlow is an **open-source machine learning framework** developed by Google. It can train and run the deep neural networks for multiple functions. TensorFlow has two parts: **tensor** is a multidimensional array and **flow** is used to define the flow of data in operation.

Process of NCF Recommender



Develop the NCF Model

Preparation → **Encode & Split Data** → **Define NCF Model** → **Training**

Preparation

- Data Loading & Cleaning
- Library Loading: LabelEncoder, train_test_split

Encode & Split Data

- Use LabelEncoder to encode user_id and item_id to user_idx and item_idx
- Split the dataset into training and test sets using train_test_split
- Extract the number of unique users and items

- Number of users: 247,383
- Number of items: 67,873

Define NCF Model

- Define input layers for users and items
- Add embedding layers
- Flatten embeddings
- Concatenate embeddings.
- Add fully connected layers and dropouts
- Compile the model with Adam optimizer and MSE loss

Training

- Prepare training data (x_train, y_train)
- Train the model using .fit()
- Output predict value using .predict()

Evaluate the NCF Model

Metric used for evaluation: Root Mean Squared Error (RMSE)

RMSE measures the average magnitude of error between predicted values (y_pred) and actual values (y_test). It provides results in the same units as the predicted variable (rating scale: 1-5).

Step 1: Use the trained model to predict ratings for the test data
Step 2: Calculate RMSE = sqrt(mean_squared_error(y_test, y_pred))

Result:

- For a movie rating, the model's predictions are, on average, off by ±0.818 units.
- The relatively low RMSE value indicate that our model is not perfect but **reasonably accurate**.

RMSE = 0.8180

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Recommend Top N Movies for User

Define a Function

Function inputs: user_id, df_meta, N (default = 5)

Check User Validity

If the user ID is not in the encoder's classes, return “User not found”

Encode & Identify Unrated Items

- Encode the user ID
- Filter out all movies that the user has already rated

Prepare Input

Create arrays for the user and the unrated movies.

Predict and Return Dataframe

Use the trained NCF model to predict ratings for all unrated items

Recommend Top N Movies

- Sort items by predicted rating in descending order and convert item indices back to their original IDs
- Join predictions with metadata (e.g., title, average rating) and return the top N recommendations

Recommendation Examples

01 Valid Example 1

User ID = 10020

	item_id	predicted_rating	title	avgRating
0	172719	5.024623	Notre Dame de Paris (1998)	4.17857
1	3226	4.810836	Hellhounds on My Trail (1999)	4.16667
2	90592	4.748853	How to Die in Oregon (2011)	4.10145
3	83560	4.733185	Macario (1960)	3.95238
4	86237	4.725561	Connections (1978)	3.79091

02 Valid Example 2

User ID = 997206

	item_id	predicted_rating	title	avgRating
0	172719	4.958755	Notre Dame de Paris (1998)	4.17857
1	142871	4.908478	O Pátio das Cantigas (1942)	4.46296
2	184393	4.707923	The Criminal Excellency Fund (2018)	4.56250
3	189159	4.677622	Born in the USSR: 14 Up (1998)	4.12500
4	3226	4.665954	Hellhounds on My Trail (1999)	4.16667

03 Invalid Example

User ID = 10000

```
Top 5 recommendations for 10000:
User not found.
```

04

Next Steps

Limitations and Recommendations

Limitations – NCF Model

01 Like most collaborative filtering models, the NCF model struggles with the **cold start problem**, where it cannot effectively make predictions for new users or items that have limited interaction data.

02 Our current model could be too generalized – it generated **similar recommendations for different users**, which could be due to the sparse user-item interactions.

The NCF model may require **more detailed tuning and refinement to improve recommendation accuracy**.

NCF Model Improvements

01 Integrate **additional layers and expand the embedding dimensions** within the NCF model. Capture **more detailed and complex patterns** in user-item interactions.

02 Combine the NCF model with **content-based filtering techniques** to create a **hybrid recommendation system.** Integrate user and item metadata and better handle new users or items by leveraging content attributes. Finally, make initial recommendations based on available content features before enough interaction data accumulates.

Other Possible Model Accuracy Evaluation Metrics

1. **Mean Absolute Error (MAE)**
- MAE treats all errors equally, giving a more direct, **linear measurement** of the average error magnitude
- A fair assessment when large errors are **not specifically more critical**
- MAE's measure is in the **same units as the output variable** (e.g., rating points), making it easier for stakeholders to understand

2. **Precision at K (Precision@K)**
- Assesses whether the **top K items** recommended by the system are actually relevant to the user, which is often more aligned with the operational goals of recommendation systems
- **High Precision@K** indicates that the recommended items are likely to be of interest to the user

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Thank you!

Any Question?

Appendix

Movie Recommendation System Python Codes

https://drive.google.com/drive/folders/1VUPnlTccyO0hwWovRHBUYNL-alSioYVW?usp=sharing